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Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation

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Introduction & Contributions

As connected automated vehicles (CAVs) enter conventional traffic streams, mixed traffic consisting of CAVs and human-driven vehicles (HDVs) is becoming increasingly common [1]. However, the benefits of CAV cooperation can be reduced by communication loss, sensor limitations, or interactions with HDVs. When a CAV follows an HDV or loses cooperative information, it may degrade to an ACC-like state, referred to as a degraded CAV (DCAV) [2][3]. Ignoring this functional degradation can lead to overestimated traffic capacity and stability. To address this issue, this study develops a unified **fundamental diagram (FD)** incorporating CAV, DCAV, and HDV interactions, derives an analytical fundamental diagram, and validates the FD model using SUMO simulations.

Main contributions:

1. Functional degradation is explicitly represented in mixed CAV-HDV traffic.
2. An analytical FD is derived through equivalent headway aggregation.
3. SUMO validation confirms the model accuracy under different CAV penetration rates.

Functional Degradation Mechanism

CAV cooperation depends on both the preceding vehicle type and communication availability. Under reliable CAV-CAV communication, the follower operates in cooperative CACC mode with two-way perception [2]. When a CAV follows an HDV or loses cooperative information, it degrades into a DCAV with ACC-like one-way sensing (see Fig. 1). This degradation increases the equivalent safe headway and weakens the capacity benefit of CAV deployment.

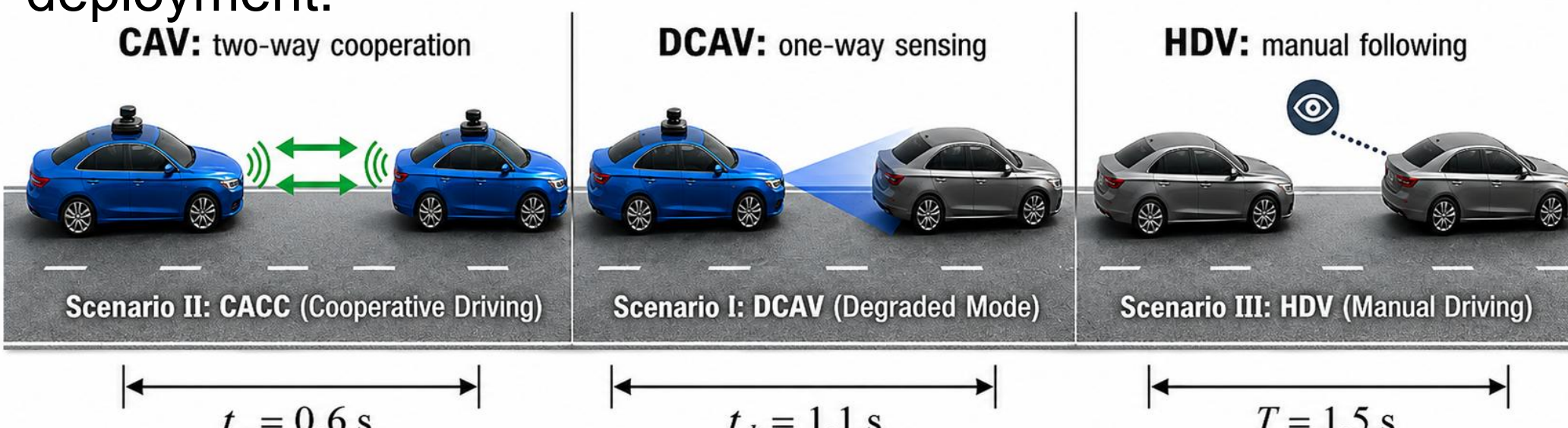


Figure 1. Vehicle interaction modes and functional degradation in mixed traffic

Mixed-Traffic Car-Following Model

The steady-state spacing of three vehicle states (see Tab. 1) is aggregated into an **expected mixed-traffic headway**, forming the basis of the analytical FD [4].

Table 1. Car-following modes and probability proportions in mixed traffic

State	Mode	Headway	Role
CAV	CACC	0.6s	cooperative following
DCAV	ACC-like	1.1s	degraded following
HDV	IDM	1.5s	manual following

Model bridge:

States → Expected headway → Analytical FD
Microscopic behavior → Macroscopic capacity

Analytical Fundamental Diagram

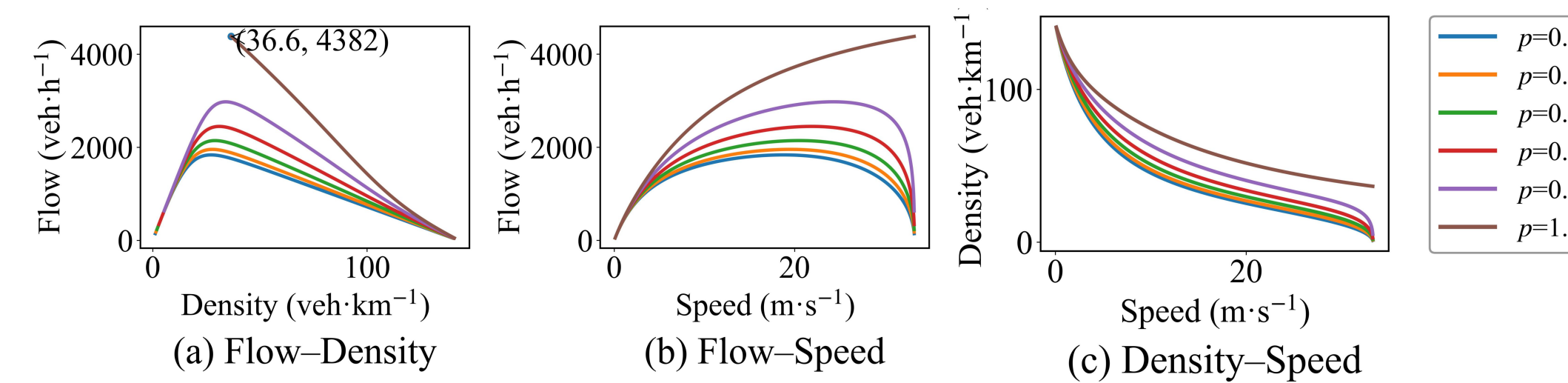


Fig. 2. Analytical FD relationships under different CAV penetration rates

See Fig. 2, the expected mixed-traffic headway is derived from CAV, DCAV, and HDV spacing. Since cooperation availability depends on density and speed, the FD becomes **implicit and nonlinear** [1][4]. **Higher CAV penetration** compresses equivalent headways and shifts the critical speed upward.

Capacity Sensitivity

Table 2. Traffic capacity indicators and sensitivity analysis under different penetrations

p	Max flow (veh/h)	Optimal density (veh/km)	Critical speed (m/s)	Marginal growth (%)	Cumulative growth (%)
0.0	1774.9	30.2	16.3	—	—
0.2	1888.5	30.3	17.3	6.4	6.4
0.4	2070.0	31.9	18.0	9.61	16.63
0.6	2361.8	33.9	19.3	14.1	33.07
0.8	2874.0	36.8	21.7	21.69	61.92
1.0	4421.1	37.6	32.7	53.83	149.09

The analytical results reveal a nonlinear capacity growth pattern (Tab. 2). At low CAV penetration, capacity gains are constrained by HDV and DCAV headways. When p exceeds approximately **0.6**, cooperative CAV interactions become dominant, leading to **rapid capacity improvement** and higher critical speed.

KEY FINDING

Capacity +149.1%

1774.9 → 4421.1 veh/h as p increases from 0 to 1

Simulation Validation

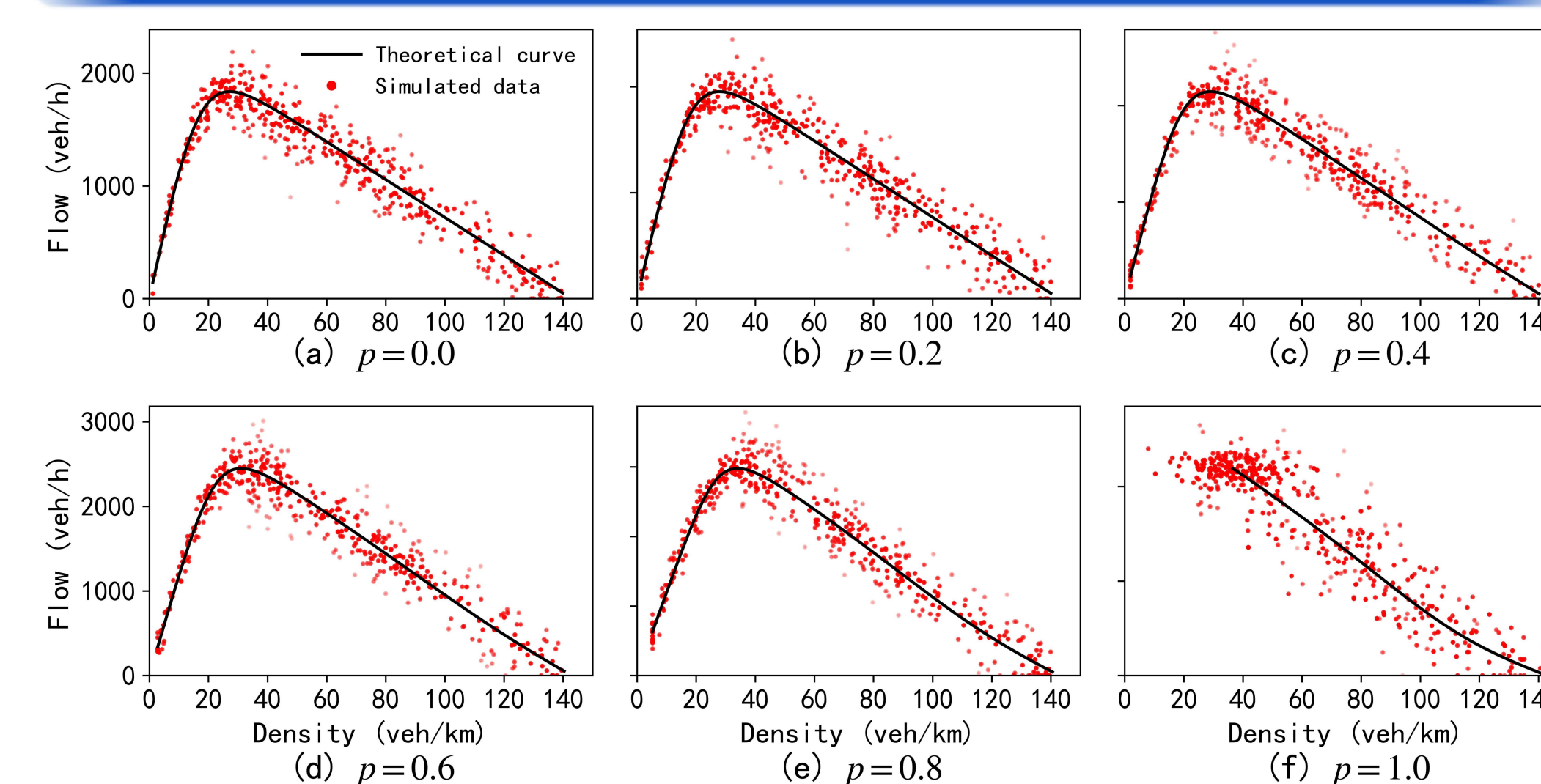


Figure 3. Comparison between simulation results and analytical FD model

As shown in Fig. 3. The simulated scatter points **closely match** the analytical FD curves, validating the proposed mixed-traffic FD model (see Table 3).

Table 3. Goodness-of-fit metrics between simulated data and analytical FD at different p

p	RMSE	R^2	$\epsilon_Q(\%)$	(%)	BCR@±5% (%)
0.0	142.6	0.963	2.1	2.4	86.3
0.2	131.4	0.969	1.9	2.2	88.7
0.4	118.1	0.975	1.6	1.9	90.9
0.6	103.7	0.981	1.4	1.7	92.8
0.8	88.9	0.986	1.2	1.5	95.1
1.0	127.5	0.972	1.8	2.1	93.0

Experimental Scenario

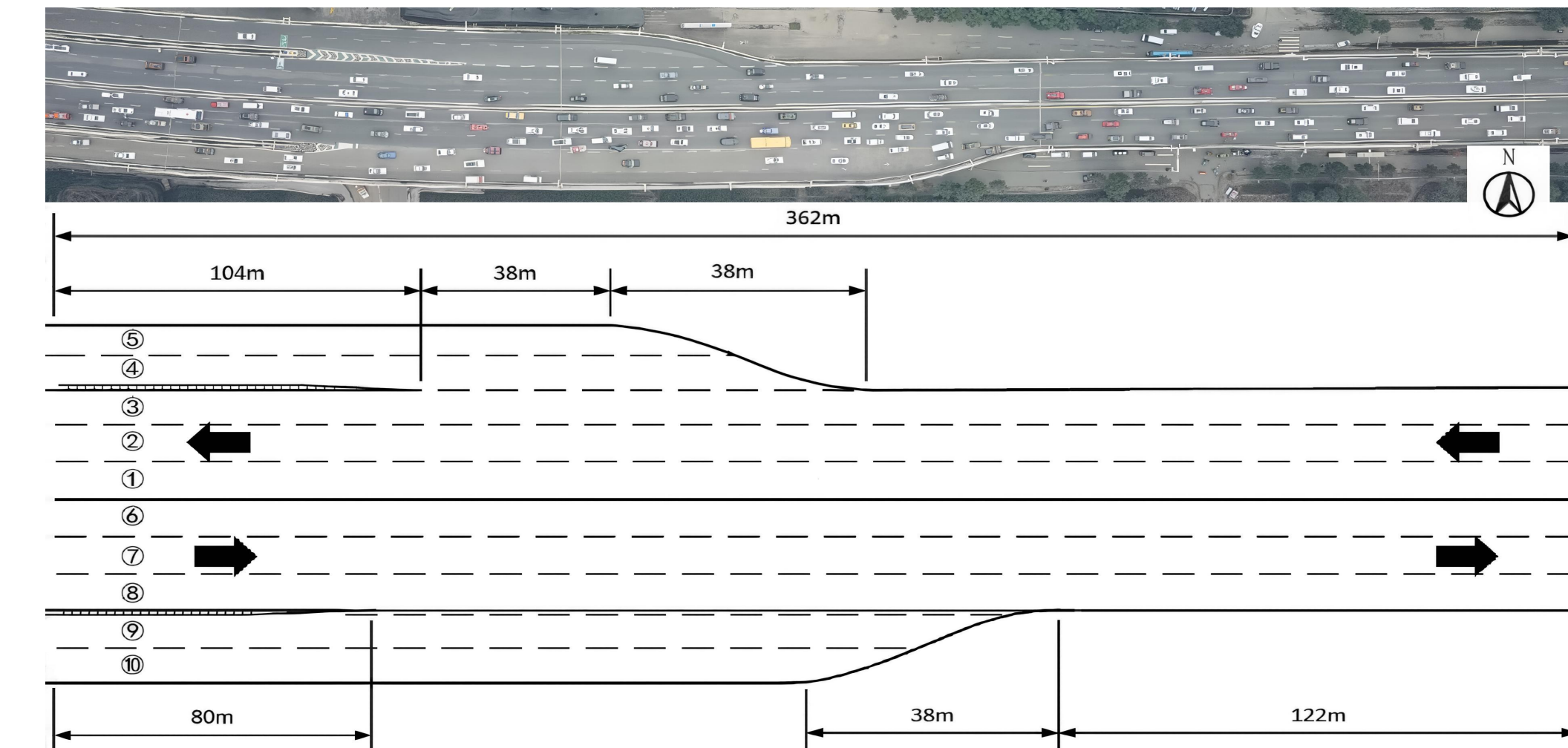


Figure 4. Aerial view and geometric layout of the expressway segment

The simulation scenario replicates a 362-meter urban expressway merging segment (Fig. 4) from the **UTE dataset** using **SUMO**, with lane-specific detectors and stochastic perturbations calibrated to real-world trajectories.

Conclusion

This study develops an analytical FD model for mixed traffic considering **CAV functional degradation**. By incorporating CAV, DCAV, and HDV behaviors into a unified headway-based formulation, the model links microscopic cooperation loss to macroscopic capacity variation. Results show that functional degradation significantly affects capacity estimation, especially when cooperative and degraded states coexist. SUMO validation confirms that the proposed FD model accurately reproduces mixed-traffic patterns, with **peak errors generally within 1–2%**. The work provides an interpretable basis for **realistic CAV benefit evaluation** and capacity assessment.

References

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Feel free to reach out and connect!

Research Interests:

Mixed Autonomy Dynamics; Urban Mobility Modelling; Multi-agent System Control.

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